

Spatial Series Analysis of Horizontal Cores to Characterize Tracer Patterns in Soil Profiles

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ABSTRACT

Surface topography, plant canopy, plant root distribution, and soil preferential flow paths cause variability in downward movement of agrichemicals in soil. To better understand this variability, tracers are often applied to the soil surface and their signatures (concentration patterns) are examined. To characterize Br^- signatures within a Crete silt loam soil (fine, montmorillonitic, mesic Pachic Argustoll), potassium bromide was applied to the soil surface at 200 kg ha^{-1} . Four sets (two sets in June and two sets in August) of four 6.1-m horizontal cores were taken in a vertical plane, perpendicular to corn plant rows at 20-, 40-, 80-, and 120-cm depths. Each 6.1-m core was divided into 120 5.1-cm sections and analyzed for Br^- . Spatial series analyses were used to identify deterministic and combined deterministic-stochastic models that described the cyclic nature of Br^- concentrations at each depth. Periods for these sinusoidal cycles generally corresponded to the distance between plant rows or the distance between the wheel tracks of field equipment.

MODERN AGRICULTURE has become very dependent on fertilizers and pesticides. Some of these agrichemicals are appearing in groundwater in unexpected amounts and times. This phenomenon is thought to be due to variation in downward movement among vertical regions in the soil profile. This variation in agrichemical movement is directly related to the volume of water passing through that region of the soil profile (Edwards et al., 1992), which, in turn, is greatly influenced by a combination of surface soil microrelief (Trojan and Linden, 1992), plant canopy (Paltineanu et al., 1995), plant root distribution (Van Wessenbeeck and Kachanoski, 1988), and preferential flow paths within the soil profile (Luxmoore, 1991). An understanding of these variations in agrichemical movement is needed so technologies and strategies can be developed to better protect groundwater quality.

Variations in agrichemical movement are revealed by examining chemical signatures (concentration patterns in the soil profile) of tracers applied to the soil surface. These signatures have been obtained from laboratory or greenhouse studies using columns and undisturbed soil blocks. Although valuable information has been obtained from these studies, field-scale chemical signatures have provided more information on the spatial variability of flow paths. Unfortunately, trends in chemical signatures (particularly in the horizontal direction) may not be readily apparent from examination of the

raw data often obtained from field-scale studies. Timlin et al. (1992) used *t*-tests for mean comparisons of data from soil samples (10 cm apart) to show differences between regions of Br^- concentration. However, the interest of this study is to investigate the dynamics of tracer concentration changes within horizontal soil cores. Spectral analysis and spatial series analysis of data from horizontally continuous samples could enhance interpretation of concentration trends in tracer signatures.

Spectral and spatial analysis has been used to characterize the effects of microtopography on soil horizon thickness and density (Kachanoski et al., 1985a,b). The objective of this study was to use spectral and spatial analysis to detect horizontal trends in Br^- concentrations that might be used to infer Br^- movement patterns beneath a corn (*Zea mays* L.) field.

METHODS AND MATERIALS

Potassium bromide (200 kg ha^{-1}) was sprayed as an aqueous solution on the soil surface of a Crete silt loam soil and the area was planted to corn on 5 May 1994. The field site was equipped with a lateral move irrigation system that restricted implement traffic to the same rows every season, resulting in alternating wheel track (W) and nonwheel track (N) interrows 51 cm wide and plant rows (R) 25 cm wide. The R and W spacings were 76.2 and 152.4 cm, center-to-center, respectively. Undisturbed, horizontal soil cores were taken by using a hydraulic press to push a core sampler (5.6 cm in diameter) into the wall of a pit dug next to the plot area. The 6.1-m-long soil cores were taken perpendicular to the plant rows at 20-, 40-, 80-, and 120-cm depths. Four sets of cores, each set in a vertical plane, were taken 5 m apart for a total of 16 cores on 20 June, 27 June, 25 August, and 26 August. A more complete description of this sampling procedure is given in Verhoeff et al. (1997), including the techniques used for keeping each set of four bore holes parallel to the soil surface and in a vertical plane.

The original intent of the experimental design was to take duplicate sets of four cores as close as possible to the same date in June and in August. This was accomplished in August; however, unexpected rain the night of 20 June and throughout the week increased the days between duplicate samplings in June to 7 d. The duplicate sets of cores for August were taken on 25 and 26 August. Total rainfall plus irrigation between Br^- application and sampling dates was 10.7, 14.7, and 47.2 cm for 20 June, 27 June, and 25/26 August, respectively. The starting point for cores taken 27 June, 25 August, and 26 August was shifted from that of 20 June to below the center of the interrow (N) because elevations of the interrows were more uniform. This allowed more precise measurements of the horizontal core depths. Each core was divided into 120 increments 5.1 cm long. Ion chromatography (Smith et al., 1995) was used to analyze each soil increment for Br^- . The limit of quantitative measurement was 0.3 mg kg^{-1} .

Preliminary examination of the data suggested that Br^-

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concentrations in the field might be cyclic with respect to horizontal distance. A more detailed statistical examination of data was performed for each core using the following four steps: (i) spectral analysis was used to select cycle periods for significance testing; (ii) a deterministic model was constructed and used to test these selected periods for significant amplitudes and phases; (iii) residuals from the deterministic model were used to construct stochastic models to identify additional periods and autocorrelations; and (iv) the deterministic model and the stochastic model were combined to describe Br^- concentration data. A detailed description of each step follows.

Spectral Analysis

Spectral analysis (Priestley, 1981) utilizing Proc Spectra (SAS Institute, 1993) was used to determine spectral density functions. From graphs of these functions, frequencies associated with maximums in spectral density (those frequencies thought to have significant spectral power) were selected and converted to periods for significance testing.

Deterministic Model Selection

With Proc Autoreg (SAS Institute, 1993), auto-regression analysis was used to examine the selected periods for statistical significance of their amplitudes and phases. The deterministic model used for this analysis is given by

$$C_j = \mu + A \cos [(2\pi\Delta_j/P) - \beta] \quad [1]$$

where C_j = concentration of Br^- in j th core increment, μ = average Br^- concentration of horizontal soil core, A = amplitude of the Cos curve (peak Br^- concentration), Δ_j = distance to the center of the j th core increment calculated as $[(5.1j) - 2.54]$ cm, P = period of Cos curve, cm, and β = phase angle of the Cos curve in radians.

Letting ϵ_j be the j th residual (difference between observed C_j and C_j predicted from Eq. [1]) and using the relationship for the cosine of the difference between two angles G and H given by

$$\cos (G - H) = \cos G \cos H + \sin G \sin H$$

in Eq. [1], the sinusoidal behavior of the data can be further specified by

$$C_j = \mu + b_1 \cos (2\pi\Delta_j/P) + b_2 \sin (2\pi\Delta_j/P) + \epsilon_j \quad [2]$$

where $b_1 = A \cos \beta$, $b_2 = A \sin \beta$, and P = periods. Additional cos and sin terms are included when more than one period was tested. The coefficients b_1 and b_2 were determined using Proc Autoreg (SAS Institute, 1993). The parameters A and β of Eq. [1] and [2] are given by Bloomfield (1976, p. 11) as $(b_1^2 + b_2^2)^{1/2}$ and $\tan^{-1} (b_2/b_1)$.

Stochastic Model Selection

The residuals from the deterministic model, ϵ_j of Eq. [2], were used to examine a portfolio of ARMA models. Based on this examination, a stochastic model was selected for each horizontal core. For the ϵ_j in Eq. [2] to meet the usual assumptions of white noise, the observed serially correlated errors, u_j , are identified by an ARMA process (Priestley, 1981). This stochastic process is given by

$$\epsilon_j = \phi(B)u_j - \theta(B)e_j \quad [3]$$

where $\phi(B)$ = backshift autoregressive operator given by

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p \quad [4]$$

$\theta(B)$ = backshift moving average operator given by

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q \quad [5]$$

and e_j = residual between u_j and ϵ_j .

The coefficients ϕ_i of $\phi(B)$ in Eq. [4] and θ_i of $\theta(B)$ in Eq. [5] were determined using ARMA techniques available from Proc ARIMA in SAS (SAS Institute, 1993). The Schwartz-Bayesian Criterion, or SBC described in Schwartz (1978), was used to select the model orders, p and q , i.e. the number of terms in $\phi(B)$ and $\theta(B)$ respectively, using the standard portfolio with neighborhood analysis as described by Parkhurst (1992). The model orders ranged from (1,0) to (12,11). However, because the moving average term, $\theta(B)$, was found to be insignificant, the ARMA model became just an AR model. The degrees of freedom were a minimum of 118 depending on the model used.

The dynamics of an AR model can be described in terms of past residuals by Green's function (Pandit and Wu, 1983). Green's function can be written as a function of the characteristic roots of the autoregressive operator. The real roots represent constant, growth, or decay trends, and the complex roots represent sinusoids with the period given by the imaginary part. Green's function describes the decay rate of the dynamic response of the system to any particular residual (or disturbance).

Combined Deterministic-Stochastic Model

Appending the right side of Eq. [3] to Eq. [1] yields a combined deterministic-stochastic model given by

$$C_j = \mu + A \cos [(2\pi\Delta_j/d) - \beta] + \phi(B)u_j - \theta(B)e_j \quad [6]$$

This combined model was used to describe the variation in Br^- concentration along the horizontal cores.

RESULTS AND DISCUSSION

Figure 1 is an example spectral density graph of the core taken on 25 August at the 20-cm depth (designated as core 20 Aug 25). In Fig. 1, prominent maximums occur at frequencies of 0.042, 0.084, and 0.157 cycles cm^{-1} . When converted to periods ($2\pi/\text{frequency}$), these frequencies are equivalent to periods of 150, 75, and 40 cm, which suggests that periods equivalent to W (152.4 cm), R (76.2 cm), and R/2 (38.1 cm) spacings be tested

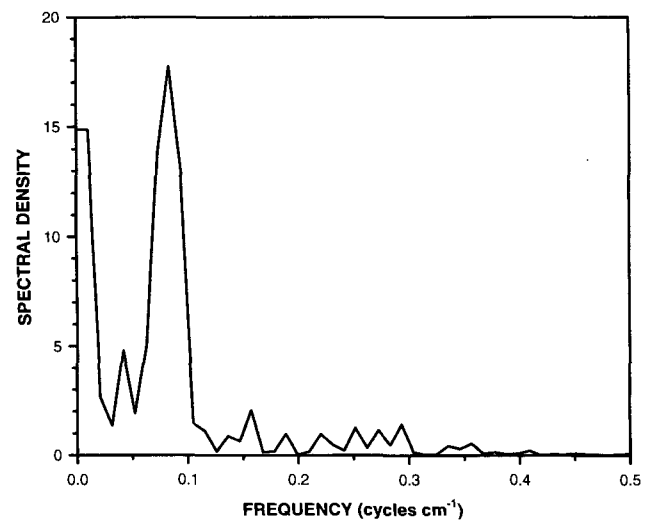


Fig. 1. Spectral density function of a horizontal core taken 25 Aug. 1994 at a depth of 20 cm.

Table 1. Average Br⁻ concentrations μ , autoregressive (AR) model coefficients (ϕ_1 and ϕ_2), wave periods for the deterministic (Detr) and stochastic (Stoch) portions of the combined (Comb) model ($P < 0.1$), and r^2 of the Detr and Comb models.

Core†	μ	AR coefficients		Wave period‡		r^2
		ϕ_1	ϕ_2	Detr	Stoch	
	mg kg ⁻¹				cm	
20 Jun 20	3.43	0.864	-0.325			
40 Jun 20	2.03	0.684			35.2	0.44
80 Jun 20	1.19	0.584		R, W		0.16
120 Jun 20	0.60	0.576	0.237	W	-§	0.04
20 Jun 27	5.20	0.876				0.05
40 Jun 27	4.05	0.847		R		0.06
80 Jun 27	2.59	1.334	-0.459	R, W		0.11
120 Jun 27	0.52	1.048	-0.375	W		0.02
20 Aug 25	3.37	1.107	-0.369	R, W	43.6	0.06
40 Aug 25	2.12	0.608		R, W		0.01
80 Aug 25	1.35	0.797		R, W		0.23
120 Aug 25	0.56	0.612	0.250	W	-	0.12
20 Aug 26	5.48	1.120		R, W	45.4	0.06
40 Aug 26	1.76	0.883	-0.445	R		0.08
80 Aug 26	0.78	0.928		R, W		0.19
120 Aug 26	0.12	-¶		R		0.04
						0.05
						-¶

† Depth (cm), month, and day of core sampling.

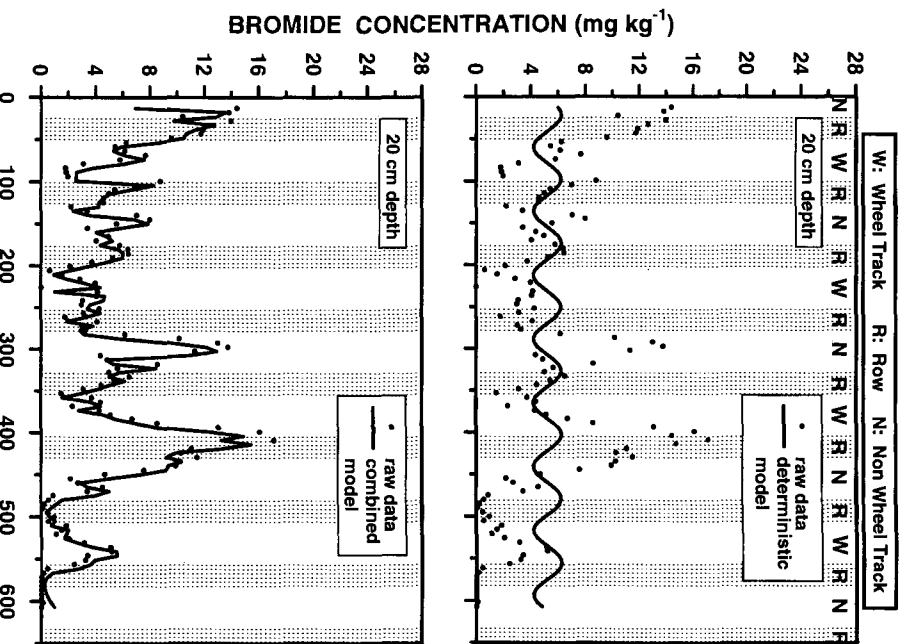
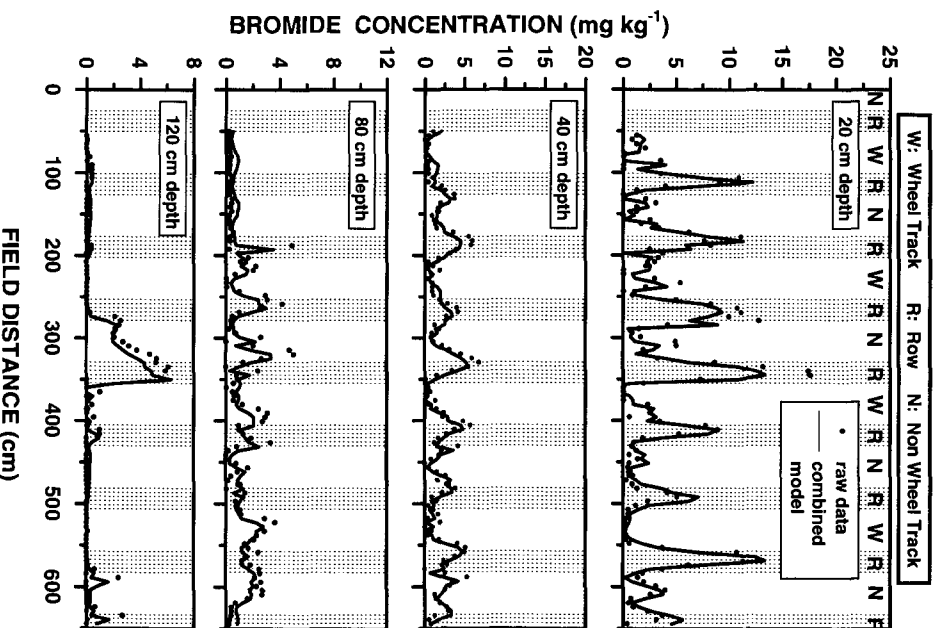
‡ R = distance between plant rows (76.2 cm); W = distance between wheel tracks (152.4 cm).

§ No complex roots to AR model, only real roots.

¶ No significant stochastic component for the model.

for significant harmonics. After repeating this process for each core and examining the resulting array of periods, it was decided to test periods equivalent to W, R, R/2, and 3R/4 for significant harmonics in all 16

horizontal cores. From Table 1, it is seen that for our example core (20 Aug 25), only periods W and R were found to have significant ($P < 0.1$) harmonics. Eq. [1] alone provides a significant deterministic

**Fig. 2.** Measured and modeled (deterministic model, Eq. [1] and combined model, Eq. [6]) Br⁻ concentrations in a soil core sampled at 20-cm depth on 27 June 1994.**Fig. 3.** Measured and modeled (Eq. [6]) Br⁻ concentrations in 20 June 1994 soil cores from 20-, 40-, 80-, and 120-cm depths.

model ($P < 0.10$) that represents the sinusoidal nature of Br^- concentration in all 16 cores. Of these, significant amplitudes and phases occurred for periods of R in 13 cores and W in 11. Both R and W spacings occurred as periods for the cosine curve in eight cores (Table 1). For a few cores, additional periods of $R/2$ or $3R/4$, or both, were found to have significant harmonics.

Although the sinusoidal nature of the data could be represented by the deterministic model in all 16 cores, it is seen from the r^2 values of Table 1 that this model accounted for less than half of the variation in Br^- concentration of the cores. At the 120-cm depth, the deterministic model accounted for $<6\%$ of the variation. Except for the 80 June 20 and 120 Aug 26 cores, the combined model accounted for 59 to 88% of the Br^- variation in the cores. The consistently smaller r^2 values for the deterministic model compared with the combined model, particularly for the two lower cores, suggest that the data are mostly stochastic at these depths. Except for the change in the 20-cm depth from the first sampling date, there appears to be no obvious trends in the partitioning of the r^2 values between the deterministic and combined models as the growing season progressed.

Periods of R and W might be explained by the effects of cyclic patterns in canopy cover, microrelief, or physi-

cal properties of the top soil caused by plant row spacings or field equipment. However, it is difficult to attribute physical meaning to periods that are a fraction of plant row spacings ($R/2$ and $3R/4$), except to note that these periods represent subharmonics of the fundamental frequencies associated with the row. When more than one period was significant for the cosine curve, the curves were added graphically to produce a single curve.

Figure 2 shows deterministic (Eq. [1]) and combined (Eq. [6]) models superimposed on measured concentration data from the core sampled at the 20-cm depth on 27 June. Here only one Cos curve (having a period of R) represents the deterministic sinusoidal nature of the data. As seen from Fig. 2 and the greater r^2 values in Table 1, the combined deterministic-stochastic model provides a better description of the Br^- signatures than the deterministic model alone. This was the case for all but one core (120 Aug 26). Green's function is a dampened sinusoidal wave of 44 to 48 cm approximating a period of $R/2$. For core 80 Jun 27, the stochastic portion of the model detected a dampened wave with a period of 74.8 cm, similar to periods of R (76.2 cm) detected in several other cores by the deterministic model.

The curves in Fig. 3 and 4 were drawn using the

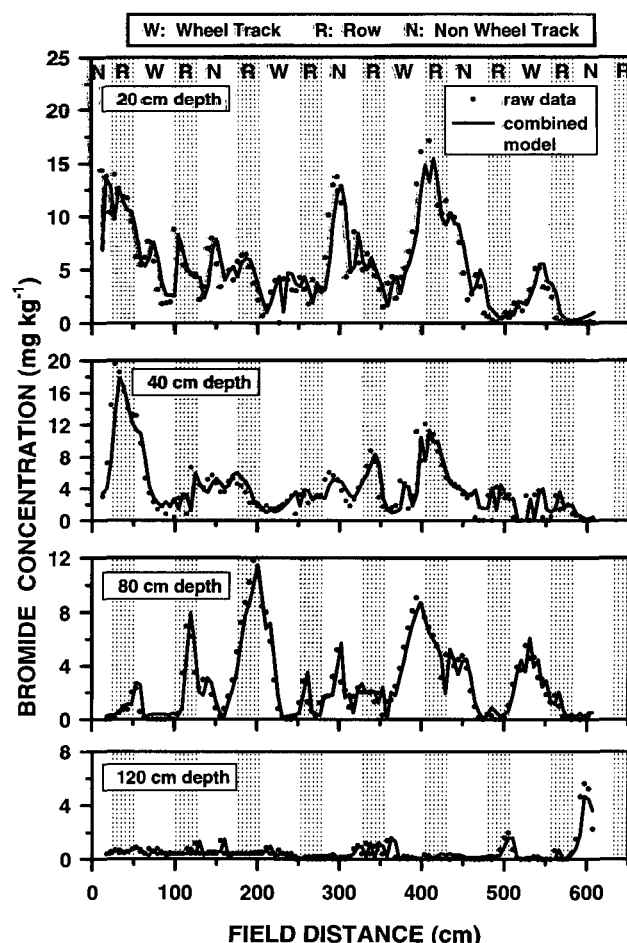


Fig. 4. Measured and modeled (Eq. [6]) Br^- concentrations in 27 June 1994 soil cores from 20-, 40-, 80-, and 120-cm depths.

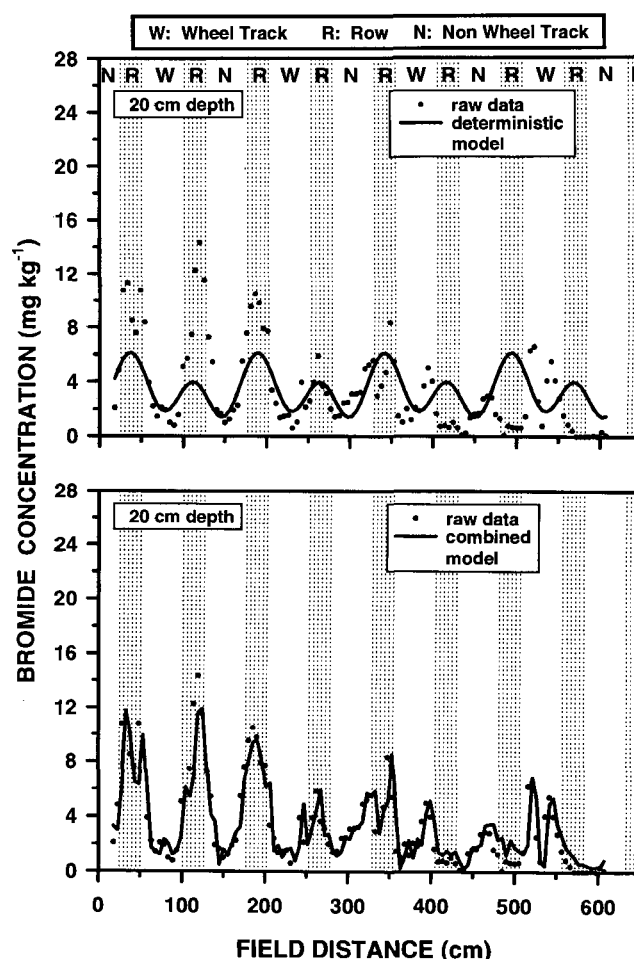


Fig. 5. Measured and modeled (deterministic model, Eq. [1] and combined model, Eq. [6]) Br^- concentrations in a soil core sampled at 20-cm depth on 25 Aug. 1994.

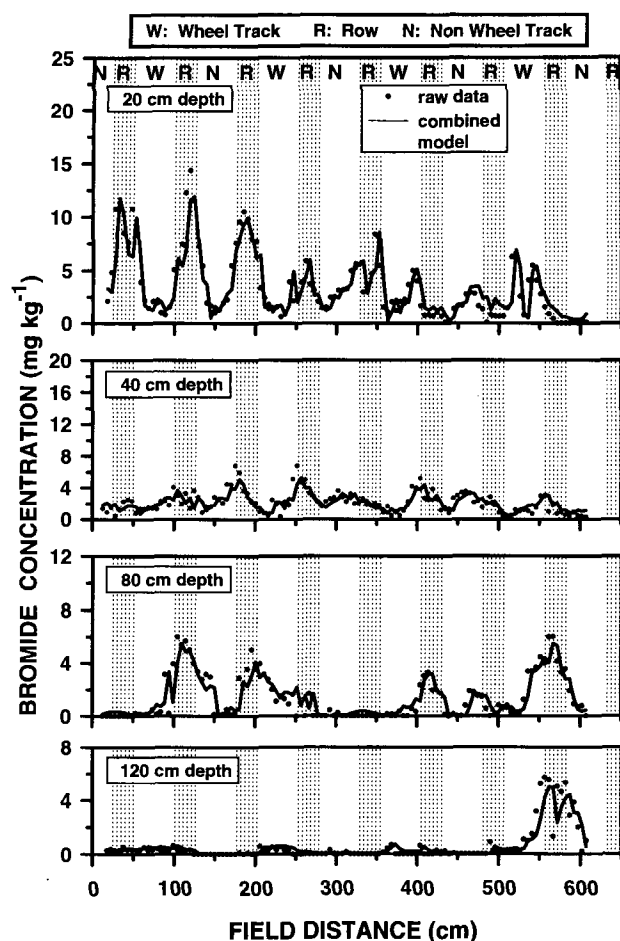


Fig. 6. Measured and modeled (Eq. [6]) Br^- concentrations in 25 Aug. 1994 soil cores from 20-, 40-, 80-, and 120-cm depths.

combined model to represent the distribution of Br^- in all cores sampled on 20 and 27 June. A majority ($\approx 70\%$) of the larger concentration maximums of these curves appear to be below the plant rows, R (more so for just the 20 June sampling date, 77%).

Curves in Fig. 5 show deterministic and combined models superimposed on measured concentration data from the core sampled at the 20-cm depth on 25 August. For the deterministic model, two cosine curves (one using R as the period and the other using W) were graphically added to represent the sinusoidal nature of the data. Again, the combined model best described the Br^- signatures by incorporating a period of ≈ 45 cm.

The curves in Fig. 6 and 7 were drawn using the combined model to represent the distribution of Br^- in all cores sampled on 25 and 26 August. As was the case for the June sampling dates, a majority ($\approx 65\%$) of the larger maximums appear below the plant rows for the first sampling date. However, for the second sampling date (26 August), the trend appears reversed and a majority (70%) of the larger maximums appear below the interrows. The striking difference between the location of the maximums from the 26 August sampling date compared with those from the other three sampling dates is difficult to explain. This difference could be due to a significant difference in microrelief between the

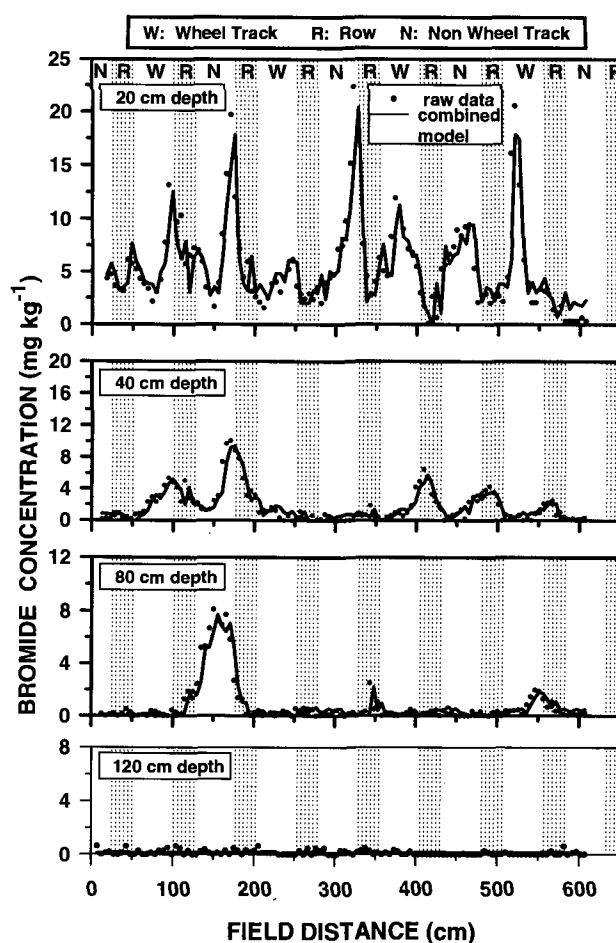


Fig. 7. Measured and modeled (Eq. [6]) Br^- concentrations in 26 Aug. 1994 soil cores from 20-, 40-, 80-, and 120-cm depths.

two sampling sites, an incorrect starting point for the set of horizontal cores in relation to the plant rows, or the fact that row spacing was not 76.2 cm between adjacent passes of the four-row planter. A 10-cm displacement in the starting point for sampling or the planter would shift some borderline Br^- concentration maximums under plant row or interrow regions.

If data from all sampling dates are used, only a slight majority (62%) of the maximums in Br^- concentration are below plant rows. Timlin et al. (1992) found higher concentrations of Br^- under corn rows than under interrows of a fine sandy loam soil. However, no such significant differences could be found in a fine sandy clay loam. For all sampling depths in this study, maximums of curves at increasing depths do not always occur directly below one another, indicating flow paths may not be strictly vertical. As expected, both the amplitudes of curves and the average Br^- concentrations appear to decrease with depth for most cases.

These statistical analyses indicate that more Br^- was removed from, did not enter, or was diverted from the regions of minimum Br^- concentration. A time sequence of Br^- concentrations in the surface soil or more frequent sets of horizontal cores is needed early in the growing season to determine which interpretation is correct. Timlin et al. (1992) used Br^- concentrations in

surface soils to conclude that Br^- had been more readily leached from regions of minimum Br^- concentrations (below interrows) in the fine sandy loam soil. However, for a loam soil, Roth et al. (1991) concluded that Cl^- had been more readily leached below regions of maximum Cl^- concentration.

CONCLUSIONS

Spectral and spatial series analyses of Br^- concentrations in horizontal cores is a useful tool in characterizing tracer signatures. However, to obtain a picture of dominant tracer pathways below corn, these analyses need to be coupled with a time sequence of Br^- concentration data from the top 10 to 15 cm of the soil profile. To use the technique described here, great care must be taken to start all horizontal cores at exactly the same location relative to the plant row and to operate the planter so that all plant row spacings are the same, particularly between passes of the planter.

Spectral analysis of data from sets of horizontal soil cores was used to select cycle periods to be tested in the model for significance. In all horizontal cores, Br^- concentrations exhibited sinusoidal signatures that were significant enough ($P < 0.10$) to be represented by deterministic Cos curves. However, since most of the sinusoidal signatures were dominantly stochastic, using a combined deterministic-stochastic model greatly improved representation of observed concentrations.

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